

ECG-INTEX Intelligent ECG Extraction and Classification Framework

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Abstract—This study presents a comprehensive framework for automated cardiac condition classification directly from twelve-lead ECG images, addressing challenges such as noise, signal variance, image artifacts, and the absence of digitized waveforms. The proposed pipeline includes image preprocessing, lead detection, adaptive segmentation, waveform digitization, and the extraction of nine time-domain features for both twelve-lead ECG signals and an extended Lead II rhythm signal. A stratified ten-fold cross-validation procedure was employed to train and evaluate multiple machine learning algorithms. Tree-based ensemble models, including XGBoost, Random Forest, and LightGBM, achieved superior performance with accuracies in the range of 95–96%. A majority-voting ensemble further improved predictive accuracy to 95.7%, demonstrating high precision and stability. For single-lead rhythm classification, CatBoost, Random Forest, and LightGBM achieved accuracies between 86–88%, with the voting ensemble reaching 87.3%. These findings highlight the robustness and clinical applicability of the proposed image-based ECG classification framework, particularly in scenarios where clean digital ECG signals are unavailable. Future work will explore deep learning methods, including 1D-CNN and 2D-CNN architectures, to facilitate real-time diagnostic capabilities.

Index Terms—ECG digitization, ensemble learning, arrhythmia classification, feature extraction, ECG images, machine learning.

I. INTRODUCTION

Electrocardiogram (ECG) is a measure of cardiac activity of the heart, which takes the form of electrical waveforms. It generally takes into account the PQRST complex. An electrode arrangement is applied on the surface of the body of a subject (patient) and these waves are followed. One ECG would suffice to forecast the kind of heartbeat, which could be normal, abnormal, or myocardial infarction (MI). The interpretation of ECGs requires proper clinical knowledge and even experienced experts might face difficulties when handling noise or vast amounts of patient data. Therefore, the automatic processing of ECG is an increasingly important topic in medical studies alongside the emergence of machine learning [2].

This paper discusses the classification of ECGs focused on the identification of Normal heartbeats, Abnormal heartbeats, and MI heartbeats. We aim to develop a fast and lightweight system that can help healthcare professionals deliver rapid and precise outcomes. Preprocessing techniques are used so that

our image-based data is converted into CSV files, followed by feature extraction performed on this CSV-based data. Finally, classification algorithms are applied to recognize the type of heartbeat. This work aims to contribute to the development of the ECG image analysis field by providing a feasible, artifact-resistant pipeline that is applicable to real-life scenarios in which clean digital signals cannot always be retrieved. This helps build scalable diagnostic tools that can assist clinicians, lessen the effort in manual interpretation, and increase accessibility in resource-restricted settings.

The remainder of the paper is organized as follows: Section II presents the literature review. Section III describes the proposed methodology. Section IV reports the experimental results. Section V concludes the paper.

II. LITERATURE REVIEW

A. ECG-Image-Database: A Large-Scale ECG Image Dataset With Real-World Artifacts (Reyna et al., 2024)

This work introduces ECG-Image-Database, a large-scale ECG image dataset containing 35,595 images generated from raw ECG time-series data using the open-source Python-based ECG-Image-Kit tool [19]. The images are derived from 977 12-lead ECG records from the PTB-XL database and 1000 12-lead records from the Emory Healthcare database. Realistic distortions such as wrinkles, stains, scanning artifacts, soaking, staining, molding, and varying lighting conditions are applied to generate challenging, artifact-rich ECG images paired with their ground-truth time-series.

The database addresses a key gap in cardiovascular research by enabling robust development and benchmarking of digitization and classification algorithms that must operate on real-world, imperfect ECG images. By capturing imaging conditions from clean scans to severely degraded prints, the dataset fosters the design of artifact-robust deep learning methods for ECG extraction and analysis.

B. Open-Source ECG Digitization Software and Validation (Fortune et al., CMPB 2022)

Fortune et al. propose and validate an open-source ECG digitization tool that converts scanned or printed ECGs into digital signals through a multi-stage process of preprocessing, grid detection, and signal extraction [20]. The pipeline uses image

normalization, rotation correction, cropping, thresholding, and frequency-domain grid analysis to produce digital waveforms from ECG images.

Using data from 230 patients, three beat types are compared: digitally recorded simultaneous median beats, digitally recorded asynchronous beats, and digitized asynchronous beats. The digitized signals exhibit a correlation of 0.977 with the corresponding digital ECGs, with a mean amplitude difference of approximately $9.3 \mu\text{V}$ and RMSE of $25.9 \mu\text{V}$. Vectorcardiographic metrics also show strong agreement, confirming that high-fidelity digitization is achievable. The main limitation is the inherent asynchrony of 3×4 ECG layouts, yet the tool remains a reproducible and accessible alternative to proprietary systems.

C. Detection of Cardiovascular Diseases in ECG Images Using Machine & Deep Learning (Abubaker & Babayigit, IEEE TAI 2023)

Abubaker and Babayigit compare traditional machine learning, transfer learning, and a newly proposed lightweight convolutional neural network (CNN) for classification of cardiovascular diseases from ECG images [21]. Four classes are considered: abnormal heartbeat, myocardial infarction, history of myocardial infarction, and normal subjects. Transfer learning approaches using SqueezeNet and AlexNet are benchmarked against the proposed CNN, which achieves 98.23% accuracy.

A further contribution is using the deep networks as feature extractors for classical machine learning algorithms such as SVM, KNN, Decision Trees, Random Forests, and Naive Bayes. With CNN-derived features, Naive Bayes achieves 99.79% accuracy, illustrating the benefit of hybrid ML-DL pipelines. Standardized preprocessing (gamma correction, resizing, z-score normalization) supports robust training. The results highlight the viability of image-based ECG diagnosis in resource-constrained settings.

D. Edge Detection Techniques—An Overview (Ziou & Tabbone, 1998)

Ziou and Tabbone provide a comprehensive survey of edge detection techniques relevant to ECG image processing tasks, including grid removal, waveform tracing, and artifact suppression [22]. They discuss edge types such as step edges, line edges, and junctions, and link them to physical phenomena including illumination changes, occlusions, and geometric structure.

The review covers smoothing, differentiation (gradient, Laplacian, directional derivatives), labeling, and multiscale design. Because differentiation is highly noise-sensitive, regularization with Gaussian filters, Green's functions, or cubic splines is emphasized. Although not ECG-specific, these foundations underpin many modern ECG digitization systems, including grid and trace extraction components used in subsequent work.

III. METHODOLOGY

The primary role of this section is to describe the process of transforming twelve-lead electrocardiogram (ECG)

images into a digital format, extracting features from the resulting signals, and performing automated classification. The methodology is applied to both the full 12-lead ECG and a single extended Lead II (rhythm lead). The pipeline includes: (1) ECG image preprocessing, (2) lead detection and region identification, (3) adaptive segment trimming, (4) waveform digitization, (5) feature extraction, (6) model development and validation, and (7) ensemble prediction. The images were obtained from a publicly accessible dataset hosted on Mendeley Data [1].

A. Image Preprocessing

The ECG images are first converted to grayscale and then binarized using Otsu's thresholding. This automatically selects an optimal threshold that maximizes the separation between ECG waveforms and the background, minimizing manual intervention. Grid lines are removed through morphological opening with rectangular kernels applied both horizontally and vertically.

B. Lead Detection and Region Identification

The raw ECG image is processed using horizontal projection profiles (channels). These profiles are smoothed using Gaussian filtering, followed by image sharpening to obtain a noise-reduced representation. Four distinct horizontal bands (channels) are identified. The top three channels are each segmented into four equal-width columns, yielding the 12 distinct lead regions. The fourth channel is treated as the rhythm strip (extended Lead II), providing a continuous reference lead. This algorithm ensures consistent localization of regions across images.

C. Segmentation

The resulting layout consists of 12 leads in a 3×4 grid plus an extended rhythm strip. Each region is treated as a segment of interest:

- **Non-Overlapping Segmentation:** Each channel is segmented into four equal, non-overlapping parts to reduce sensitivity to local distortions and variability.
- **Adaptive Segment Trimming:** Printed annotations (e.g., I, II, aVL, aVR) and labels at the boundaries introduce local noise. To minimize their impact, each segment undergoes adaptive trimming at its horizontal boundaries. A small fixed percentage is removed from both ends of every segment, mitigating label interference, ink thickness artifacts, and local noise, while preserving ECG morphology.

D. Waveform Digitization

Each trimmed segment is transformed from a 2D trace to a 1D digital signal representation. For each column, the median vertical coordinate of white pixels is interpreted as the signal amplitude at that time instant. Missing values are imputed via linear interpolation. Signals are inverted vertically, zero-centered, and standardized in amplitude. The resulting waveforms are stored in both `.npz` and `.csv` formats. For

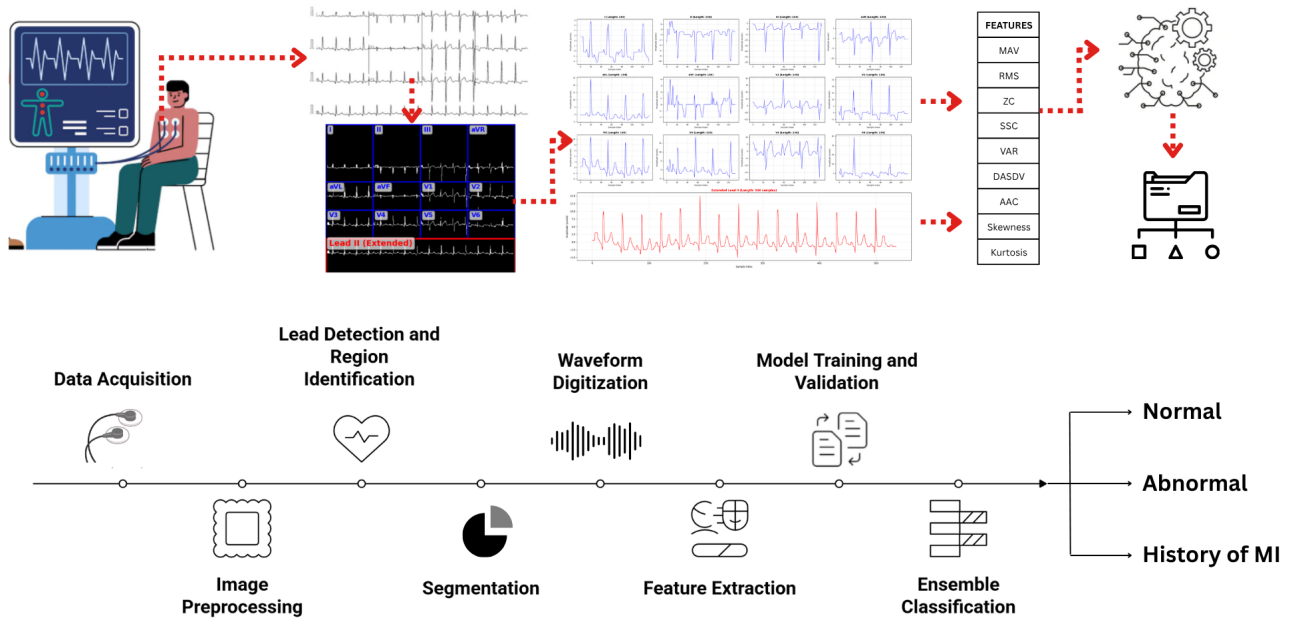


Fig. 1. Overview of the ECG-INTEX pipeline: data acquisition, image preprocessing, lead detection and region identification, segmentation, waveform digitization, feature extraction, and ensemble-based classification into normal, abnormal, and history-of-MI classes.

each ECG image, two files are generated: one for the 12-lead signals and one for the rhythm lead.

E. Feature Extraction

From each digitized ECG waveform, nine time-domain features are extracted to capture amplitude, variability, and higher-order statistical behavior.

These features are concatenated with a label-encoded class for each ECG image to form a structured dataset for model training, helping to mitigate noise and class imbalance effects.

F. Model Training and Validation

The feature dataset is used to train a range of supervised and semi-supervised models. Performance is evaluated via stratified 10-fold cross-validation to preserve class proportions in each fold and improve generalization.

The classifiers include:

- Support Vector Machines (SVM) with linear, RBF, and polynomial kernels,
- Decision Tree and Random Forest,
- Extreme Gradient Boosting (XGBoost),
- LightGBM and CatBoost,
- Gaussian Mixture Model (GMM) as a semi-supervised baseline,
- Deep Neural Network (DNN) and Multilayer Perceptron (MLP) with batch normalization and dropout.

Prior to each cross-validation fold, features are standardized using `StandardScaler`. Performance metrics include accuracy, precision, recall, F1-score, and confusion matrices. Cross-validated mean accuracy and standard deviation across

10 folds are reported for both the 12-lead and rhythm-only datasets.

G. Ensemble Classification

To further stabilize predictions, a majority voting ensemble is constructed from the top three models (based on cross-validated accuracy). For each test instance, the predicted class labels from these three models are aggregated, and the majority class is used as the final prediction. This reduces variance and improves robustness compared to single-model predictions.

IV. RESULTS AND DISCUSSION

Experimental evaluation demonstrates strong classification performance for a variety of machine learning algorithms, with ensemble methods producing the best overall results on both feature sets.

A. Standard Features Performance (12-Lead ECG)

According to Table II, XGBoost and Random Forest achieve the best individual accuracy of 0.9552, highlighting the strength of tree-based ensemble methods. The DNN model follows closely with an accuracy of 0.9532 and the lowest standard deviation (0.0234), indicating strong stability. The Voting Ensemble further improves performance to 0.9571 accuracy, demonstrating the benefits of combining complementary models.

Traditional methods show mixed results: linear SVM reaches 0.9103 accuracy, whereas GMM yields only 0.4853, suggesting that the feature distribution is not well modeled by simple Gaussian mixtures. The ensemble's high precision (0.9589) and recall (0.9571) indicate balanced performance across classes, which is crucial in clinical decision support.

TABLE I
EXTRACTED TIME-DOMAIN FEATURES FROM EACH ECG SIGNAL

Feature Name	Formula
Mean Absolute Value (MAV)	$MAV = \frac{1}{N} \sum_{i=1}^N x_i $
Root Mean Square (RMS)	$RMS = \sqrt{\frac{1}{N} \sum_{i=1}^N x_i^2}$
Zero Crossings (ZC)	$ZC = \sum_{i=1}^{N-1} \begin{cases} 1, & x_i x_{i+1} < 0, \\ 0, & \text{otherwise} \end{cases}$
Slope Sign Changes (SSC)	$SSC = \sum_{i=2}^{N-1} \begin{cases} 1, & (x_i - x_{i-1})(x_i - x_{i+1}) > 0, \\ 0, & \text{otherwise} \end{cases}$
Variance (VAR)	$VAR = \frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2$
DASDV	$DASDV = \sqrt{\frac{1}{N-1} \sum_{i=1}^{N-1} (x_{i+1} - x_i)^2}$
Average Amplitude Change (AAC)	$AAC = \frac{1}{N-1} \sum_{i=1}^{N-1} x_{i+1} - x_i $
Skewness	$Skewness = \frac{1}{N} \sum_{i=1}^N \left(\frac{x_i - \bar{x}}{\sigma} \right)^3$
Kurtosis	$Kurtosis = \frac{1}{N} \sum_{i=1}^N \left(\frac{x_i - \bar{x}}{\sigma} \right)^4$

TABLE II
PERFORMANCE COMPARISON OF ML MODELS ON 12-LEAD ECG FEATURES

Model	Mean Acc.	Std Dev	Precision	Recall	F1-Score
SVM (Linear)	0.9103	0.0292	0.9141	0.9103	0.9096
SVM (RBF Kernel)	0.9043	0.0344	0.9071	0.9045	0.9049
SVM (Poly Kernel)	0.7913	0.0433	0.8521	0.7914	0.7940
XGBoost	0.9552	0.0387	0.9561	0.9552	0.9551
MLP	0.8634	0.0648	0.8641	0.8635	0.8638
Decision Tree	0.8637	0.0422	0.8688	0.8635	0.8623
Random Forest	0.9552	0.0302	0.9563	0.9552	0.9550
CatBoost	0.9454	0.0310	0.9475	0.9454	0.9454
GMM	0.4853	0.0468	0.4569	0.4854	0.4219
LightGBM	0.9474	0.0323	0.9492	0.9474	0.9473
DNN	0.9532	0.0234	0.9550	0.9532	0.9531

TABLE III
VOTING ENSEMBLE PERFORMANCE ON 12-LEAD ECG FEATURES

Model	Mean Acc.	Std Dev	Precision	Recall	F1-Score
Voting Ensemble	0.9571	0.0285	0.9589	0.9571	0.9570

Note: The Voting Ensemble combines XGBoost (0.9552), Random Forest (0.9552), and DNN (0.9532).

TABLE IV
PERFORMANCE COMPARISON OF ML MODELS ON RHYTHM (LEAD II) FEATURES

Model	Mean Acc.	Std Dev	Precision	Recall	F1-Score
SVM (Linear)	0.6275	0.0612	0.6208	0.6277	0.6144
SVM (RBF Kernel)	0.6471	0.0595	0.6487	0.6472	0.6354
SVM (Poly Kernel)	0.5495	0.0591	0.6215	0.5497	0.5271
XGBoost	0.8598	0.0521	0.8636	0.8596	0.8596
MLP	0.6063	0.0464	0.5870	0.6062	0.5871
Decision Tree	0.7917	0.0717	0.7954	0.7914	0.7916
Random Forest	0.8695	0.0546	0.8721	0.8694	0.8692
CatBoost	0.8773	0.0363	0.8802	0.8772	0.8769
GMM	0.4990	0.0547	0.5505	0.4990	0.4446
LightGBM	0.8618	0.0440	0.8642	0.8616	0.8616
DNN	0.6647	0.0688	0.6707	0.6647	0.6609

TABLE V
VOTING ENSEMBLE PERFORMANCE ON RHYTHM (LEAD II) FEATURES

Model	Mean Acc.	Std Dev	Precision	Recall	F1-Score
Voting Ensemble	0.8734	0.0605	0.8758	0.8733	0.8732

Note: The Voting Ensemble combines CatBoost (0.8773), Random Forest (0.8695), and LightGBM (0.8618).

B. Rhythm Features Performance (Single-Lead)

Table IV shows that classification based on rhythm-only features is more challenging than with full 12-lead input. The best individual performance is obtained by CatBoost with

0.8773 accuracy. The Voting Ensemble combining CatBoost, Random Forest, and LightGBM achieves 0.8734 accuracy with reasonable stability (standard deviation 0.0605). The reduction in overall performance compared to the 12-lead case indicates

that single-lead rhythm information is less discriminative, but ensembles still help mitigate variability.

C. Key Findings

Several important observations emerge:

- 1) Ensemble methods consistently outperform the best single classifiers, both for 12-lead and rhythm-only settings.
- 2) Gradient boosting algorithms (XGBoost, CatBoost, LightGBM) demonstrate robust performance across different feature configurations.
- 3) Standard (12-lead) features provide roughly 10% higher accuracy than rhythm-only features, reflecting the added information in multi-lead recordings.
- 4) Model stability, measured via standard deviation across folds, is critical in medical applications, where unreliable predictions are unacceptable.

These findings support the deployment of ensemble-based methods for automated ECG interpretation and arrhythmia classification.

V. CONCLUSION

This work presents a complete automated ECG interpretation framework that spans image preprocessing, waveform digitization, time-domain feature extraction, and ensemble-based classification. The framework is evaluated on both full 12-lead ECG signals and an extended rhythm-lead configuration, demonstrating strong generalization across input formats.

For 12-lead ECGs, ensemble models such as XGBoost, Random Forest, and LightGBM achieve accuracies in the 95–96% range, with the voting ensemble reaching 95.7%. This highlights the robustness and reduced variance provided by ensemble learning. For the rhythm-only dataset, despite relying on a single lead (Lead II), the system still achieves 86–88% accuracy with top gradient boosting models and 87.3% with the ensemble, showing that well-designed features can partially compensate for limited signal diversity.

Overall, the proposed framework is effective, scalable, and clinically promising. Future work will explore integrating 1D-CNN and 2D-CNN architectures to automatically learn richer temporal and morphological representations, aiming to further improve performance and move toward real-time diagnostic deployment.

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